DEEP LEARNING – FUNDAMENTALS OF NEURAL NETWORK

1. From Machine Learning to Deep Learning

We have already completed the basics of **Machine Learning**, including Supervised and Unsupervised Learning.

Now we are moving to **Deep Learning**, which is a part of Machine Learning.

Machine Learning

In traditional Machine Learning, we usually:

Data → Select/Prepare Features → ML Algorithm → Prediction

For example, to predict whether a student will pass, we may provide:

- Study Hours
- Attendance
- Assignment Score

The ML model learns from these features and predicts:

Pass / Fail

Deep Learning

Deep Learning uses **Neural Networks with multiple layers**.

Instead of depending heavily on manually selected features, neural networks can learn useful patterns and representations through these layers.

Data → Neural Network Layers → Prediction

Simple Example: Image Recognition

Suppose we want to identify a cat in an image.

A Deep Learning model can gradually learn:

Pixels → Edges → Shapes → Eyes/Ears → Cat

So, the important difference is:

Machine Learning: We often prepare/select useful features → Model learns → Prediction

Deep Learning: Neural Network learns representations through multiple layers → Prediction

Remember

Deep Learning is not simply Machine Learning with more layers.

It is a type of Machine Learning based on **neural networks with multiple layers that can learn increasingly complex patterns from data.**

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2. What is Deep Learning?

Deep Learning is a branch of Machine Learning that uses **neural networks with multiple layers** to learn complex patterns from data.

It is commonly used for:

• Image Recognition
• Speech Recognition
• Text Processing
• Classification
• Prediction

Example

For recognizing a face:

Image → Simple Patterns → Facial Features → Face → Prediction

Each layer can learn a different level of information.

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3. What is a Neural Network?

A Neural Network is a model made up of interconnected **neurons** that learn patterns from data.

A basic neural network contains:

Input Layer → Hidden Layer(s) → Output Layer

Example: Student Result Prediction

Suppose we want to predict whether a student will pass.

Input:

Study Hours
Attendance
Assignment Score

↓

Neural Network

↓

Output:

Pass / Fail

The network learns the relationship between the inputs and the final result.

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4. Layers of a Neural Network

Input Layer

The input layer receives the features from our dataset.

Example:

Study Hours | Attendance | Assignment Score

Hidden Layer

Hidden layers process the information and learn patterns.

For example, the network may learn:

More study + good attendance → Higher chance of passing

A neural network can contain one or many hidden layers.

More hidden layers allow the network to learn more complex patterns.

Output Layer

The output layer gives the final prediction.

Example:

0 → Fail

1 → Pass

For multi-class problems, the output can contain multiple classes.

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5. What is a Neuron?

A **neuron** is the basic building block of a neural network.

A neuron:

Takes Inputs → Applies Weights → Adds Bias → Applies Activation Function → Produces Output

Example

For student prediction:

Study Hours → Weight

Attendance → Weight

Assignment Score → Weight

The neuron combines these values and produces an output.

Basic idea:

Output = Activation(Weighted Sum + Bias)

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6. What are Weights?

A **weight represents the importance of an input**.

Example

Suppose Study Hours are very important for predicting whether a student will pass.

The neural network may learn a larger weight for Study Hours.

Another less important feature may receive a smaller weight.

Important: We do not manually decide the final weights.

The neural network **learns the weights during training**.

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7. What is Bias?

Bias is an additional value added to the weighted sum.

Basic idea:

Weighted Sum = Inputs × Weights + Bias

Bias gives the neuron additional flexibility to make better decisions.

So, a neuron works roughly like:

Inputs + Weights + Bias → Activation → Output

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8. Perceptron

A **Perceptron** is one of the simplest forms of a neural network.

It contains a single neuron.

Flow

Input → Weighted Sum + Bias → Activation → Output

Example

Study Hours + Attendance

↓

Perceptron

↓

Pass / Fail

A perceptron is mainly used for simple binary classification.

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9. Activation Functions

An **Activation Function** determines how the output of a neuron should be transformed.

Its main purpose is to introduce **non-linearity**.

Why do we need non-linearity?

Real-world problems are complex.

Activation functions help neural networks learn complex relationships instead of only simple linear relationships.

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10. Important Activation Functions

Sigmoid

Sigmoid produces values between:

0 and 1

This makes it useful for probability-like outputs.

Example

Model Output:

0.85

This can be understood as approximately:

85% probability

It is commonly used for **binary classification**.

Example:

0 → Fail

1 → Pass

ReLU

ReLU stands for **Rectified Linear Unit**.

It converts negative values to 0 and keeps positive values.

Examples:

-5 → 0

3 → 3

8 → 8

ReLU is commonly used in **hidden layers**.

Tanh

Tanh produces values between:

-1 and +1

It can be useful when zero-centered output is preferred.

Leaky ReLU

Leaky ReLU is similar to ReLU but allows a small negative value instead of converting every negative value to 0.

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11. What is a Loss Function?

A **Loss Function measures how wrong the model's prediction is.**

It compares:

Actual Value vs Predicted Value

Example

Actual = **1**

Prediction = **0.9**

→ Small error

Actual = **1**

Prediction = **0.1**

→ Large error

The goal of training is to:

Reduce the Loss

Remember:

Lower Loss → Better prediction

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12. Common Loss Functions

MSE – Mean Squared Error

Commonly used for **Regression**.

Examples:

• Salary Prediction
• House Price Prediction
• Temperature Prediction

MAE – Mean Absolute Error

Also commonly used for **Regression**.

Binary Cross-Entropy

Used for **Binary Classification**.

Examples:

- Spam / Not Spam
- Pass / Fail
- Yes / No

Categorical Cross-Entropy

Used for **Multi-Class Classification**.

Example:

Cat / Dog / Horse

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13. How Does a Neural Network Learn?

A neural network learns by:

Making a prediction → Checking the error → Correcting itself → Repeating

The complete process is:

Input Data

↓

Forward Pass

↓

Prediction

↓

Calculate Loss

↓

Backpropagation

↓

Update Weights

↓

Repeat

This repeated process allows the network to learn useful patterns.

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14. Forward Pass

During the **Forward Pass**, information moves from:

Input Layer → Hidden Layer(s) → Output Layer

The network uses its current weights and produces a prediction.

Example

Input:

Study Hours = 6
Attendance = 85%

↓

Neural Network

↓

Prediction = **0.80**

The model predicts an 80% probability of passing.

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15. Backpropagation

After making a prediction, the model compares it with the actual answer.

If there is an error, the network needs to know:

“Which weights should I change?”

Backpropagation sends the error information backward through the network and calculates how the weights should be adjusted.

Simple Understanding

Prediction → Error → Find contribution of weights → Adjust weights

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16. Gradient Descent

Gradient Descent is an optimization technique used to reduce the loss.

Think of it like trying to reach the **lowest point of a mountain**.

The model looks for the direction in which the loss decreases and adjusts the weights accordingly.

Simple Flow

High Loss

↓

Adjust Weights

↓

Lower Loss

↓

Adjust Again

↓

Even Lower Loss

This process continues during training.

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17. Neural Network vs Deep Neural Network

Neural Network

Input → Hidden Layer → Output

Deep Neural Network

Input → Hidden Layer 1 → Hidden Layer 2 → Hidden Layer 3 → Output

The multiple hidden layers allow the network to learn increasingly complex patterns.

Example: Image Recognition

Image

↓

Layer 1 → Edges

↓

Layer 2 → Shapes

↓

Layer 3 → Objects/Features

↓

Output → Cat

This is the basic idea behind Deep Learning.

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18. Complete Neural Network Learning Cycle

Remember this complete flow:

Input Data

↓

Weights + Bias

↓

Activation Function

↓

Prediction

↓

Loss

↓

Backpropagation

↓

Gradient Descent

↓

Update Weights

↓

Repeat

In One Line:

Predict → Calculate Error → Correct → Repeat

That is how a neural network learns from data.

QUICK REVISION

Machine Learning
Usually requires us to select or prepare useful features before training the model.

Deep Learning
Uses multi-layer neural networks that can learn increasingly complex representations from data.

Neural Network
A model made of interconnected neurons.

Neuron
Basic processing unit of a neural network.

Weight
Represents the importance of an input.

Bias
An additional value that gives the neuron more flexibility.

Activation Function
Introduces non-linearity and transforms the neuron's output.

Loss Function
Measures how wrong the prediction is.

Forward Pass
Moves information through the network to produce a prediction.

Backpropagation
Determines how the weights should be adjusted based on the error.

Gradient Descent
Helps reduce the loss by updating the weights.

Deep Learning
Uses neural networks with multiple layers to learn complex patterns.

KEY IDEA TO REMEMBER

Machine Learning

Features → ML Algorithm → Prediction

Deep Learning

Data → Multiple Neural Network Layers → Learned Patterns → Prediction

Neural Network Learning

Input → Prediction → Loss → Backpropagation → Weight Update → Repeat

Prediction → Error → Correction → Learning